

HOME TASK - ADVANCED SQL QUERIES

Aim

To strengthen SQL problem-solving skills by applying joins, aggregate functions, GROUP BY, HAVING, LEFT JOIN, NULL handling, and analytical queries to solve practical database scenarios.

Objectives

- Understand the use of GROUP BY and HAVING clauses.
- Apply INNER JOIN and LEFT JOIN for relational data retrieval.
- Develop SQL queries using aggregate functions.
- Identify missing or unmatched records using NULL conditions.
- Analyze business scenarios and write optimized SQL solutions.

Question 1 (Easy): Employee Certification Matching

Find employees who possess all three skills: SQL, Python, and Excel.

Solution

```
SELECT e.employee_id, e.employee_name
FROM Employees e
JOIN Employee_Skills s
ON e.employee_id=s.employee_id
WHERE s.skill IN ('SQL','Python','Excel')
GROUP BY e.employee_id,e.employee_name
HAVING COUNT(DISTINCT s.skill)=3
ORDER BY employee_id;
```

Question 2 (Medium): Products Without Reviews

Find products that have never received a review.

Solution

```
SELECT p.product_id,p.product_name
FROM Products p
LEFT JOIN Reviews r
ON p.product_id=r.product_id
WHERE r.review_id IS NULL
ORDER BY p.product_id;
```

Question 3 (Hard): Warehouse Orders Pending Completion

Find remaining quantity to be shipped for each order.

Solution

```
SELECT o.order_id,o.item_type,
o.ordered_qty-COALESCE(SUM(s.shipped_qty),0) AS remaining_quantity
FROM Orders o
```

```
LEFT JOIN Shipments s
ON o.order_id=s.order_id
GROUP BY o.order_id,o.item_type,o.ordered_qty
HAVING o.ordered_qty-COALESCE(SUM(s.shipped_qty),0)>0
ORDER BY o.order_id;
```

Learning Outcomes

- Write SQL queries using aggregate functions and grouping.
- Use JOIN operations to combine related tables.
- Retrieve missing records using LEFT JOIN and NULL filtering.
- Apply HAVING clause for grouped data filtering.
- Develop optimized SQL solutions for real-world database problems.
- Improve analytical thinking and query optimization skills.